SWAPNIL KUMAR

London, United Kingdom | swapnil.kumar22@alumni.imperial.ac.uk | 07543756454 |
https://github.com/SwapnilKumar08, https://scholar.google.com/citations?user=IYPVDYUAAAAJ&hl=en&authuser=2

SUMMARY

AI/ML engineer with years of experience delivering production AI systems, machine learning research, and technical programmes across industry and academia. Built LLM/RAG applications, semantic retrieval systems, knowledge graphs, educational platforms, and cloud automation; led a 10-person cross-functional team and worked with large-scale datasets.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, LLM integration, RAG, agentic workflows, semantic vector databases, knowledge graphs, PINNs, Bayesian optimisation

Full Stack: React 18, Vite, Tailwind CSS, Supabase, PostgreSQL, REST APIs, pandas, NumPy, SciPy, DBT, BigQuery, Selenium

Cloud / DevOps: AWS, GCP, Azure, Docker, Terraform, Ansible, GitHub Actions, CI/CD, HPC, MPI, OpenMP, Prometheus

PROFESSIONAL EXPERIENCE

A-Team Academy — Full-Stack Engineer / AI Engineer **Feb 2026 – June 2026**

- Developed two production educational platforms, EconRev and Predicted Papers, spanning personalised revision, predicted examination papers, automated marking, model answers, diagram practice, grade estimation, dashboards, CRM workflows, and payments. Architected React 18, Vite, TypeScript, Tailwind, and shadcn/ui front ends with Supabase PostgreSQL, Auth, Storage, Row-Level Security, Realtime, and Edge Functions for secure, scalable delivery.
- Integrated Claude and OpenAI models through an AI gateway, applying RAG, semantic vector search, knowledge graphs, and computer vision to improve syllabus alignment, feedback quality, and diagram verification.
- Automated subscription, customer-engagement, lead-management, and learning-feedback workflows to reduce manual operations and create end-to-end student and staff experiences.

XLAB Innovations Ltd — Director **Sep 2025 – Present**

- Defined the product and technical strategy for AI-assisted organ printing by integrating microfluidic bioprinting, physics-informed modelling, uncertainty quantification, and adaptive flow control.
- Developed surrogate modelling and physics-informed neural network architectures to accelerate patient-specific construct development, reduce simulation costs, and improve experimental decision-making.
- Implemented semantic vector databases and knowledge graphs to manage large-scale structured & unstructured datasets. Established the organization's AI strategy and product architecture, translating complex business and user requirements into scalable AI systems.
- Created workflow automation, knowledge navigation, and real-time guidance capabilities through natural-language interaction. Engineered regulatory document ingestion and processing pipelines for structured rule extraction, classification, and interpretation. Architected scalable semantic retrieval solutions to improve the accuracy and accessibility of regulatory information.
- Constructed knowledge graphs to map regulatory obligations, rule dependencies, and interconnected requirements.
- Delivered evidence-backed answers to business questions, supported by citations linked directly to source documentation. Introduced change-impact assessment processes to evaluate the effect of regulatory updates on obligations and business operations. Ensured end-to-end auditability and traceability by linking every regulatory interpretation to the source documentation.
- Establishing evaluation frameworks, governance principles, and ethical guardrails to ensure responsible, explainable, and high-impact AI deployment.
- Driving cross-functional collaboration between product, engineering, and design to embed AI capabilities into user-facing applications.

Propeterra — Artificial Intelligence Engineer **Jun 2025 – Sep 2025**

Artificial Intelligence Engineer

- Integrated multiple Large Language Models (LLMs) into a unified AI platform to improve the accuracy, relevance, and quality of data-driven responses. Designed and implemented end-to-end AI pipeline architectures, including data ingestion, model orchestration, and automated workflows.
- Conducted large-scale data collection using non-biased LLM-driven strategies and trained models using APIs and AWS cloud infrastructure. Developed Python automation scripts for large-scale business email discovery & automated communication workflows. Built a complete Selenium-based web scraping framework, including LinkedIn data extraction for professionals. Implemented automated data extraction and email outreach pipelines integrated with hunter.io. Utilised Jira and Confluence for agile project management, documentation, and cross-team collaboration.

Data Engineering & AI Systems Development

- Custom data collection framework for Commercial Real Estate properties across Asia & UK Council datasets. Performed time-series analysis and heatmap-based geospatial analytics to extract market and policy insights. Applied Retrieval-Augmented Generation (RAG) to interpret structured and unstructured data sources. Enabled automated reading and reasoning over PDF documents stored in repositories to support analytical and training workflows. Generated analytical visualisations to support downstream machine learning and decision-making processes. Collaborated extensively using Jira and Confluence to ensure reproducible, well-documented workflows.

Software Engineering, DevOps & Large-Scale Data

- Managed the organisation's GitHub infrastructure, ensuring standardised JSON-based data storage & version control. Implemented GitHub-based CI/CD pipelines for reliable, consistent & version-controlled cloud resource deployment.
- Adhered to best practices in Git/GitHub version control, testing frameworks, and CI/CD for collaborative research. Conducted software benchmarking, profiling, and code optimisation to improve system efficiency and sustainability. Worked extensively with large-scale datasets exceeding one million records.

Leadership, Product Launch & Applied Machine Learning

- Led and coordinated a cross-functional team of 10 members, including AI Engineers, Data Engineers, and Geospatial Engineers. Ensured timely delivery of technical milestones through effective planning, communication, and collaboration. Supported CEO with organisational funding strategies and product development initiatives. Designed, developed, and deployed advanced ML models in close collaboration with analysts & engineers. Explored and analysed diverse datasets to uncover actionable business and market insights. Selected, trained, and rigorously evaluated AI and machine learning models for accuracy, robustness, and performance. Ensured seamless integration of models into production environments, maintaining high standards of scalability, reliability, and efficiency.

ETH Zurich – Research Fellow (Zurich, Switzerland) **July 2022 – Sep 2022**

Research: Design, Modeling, Model predictive control, Control, Reinforcement learning & Trajectory Optimization.

Project: Design evaluation and performance optimization of an all-wheel drive miniature car capable of torque vectoring

University of Louisville – Project Engineer (Louisville, Kentucky, United States) **Aug 2020 – Sep 2022**

Design for Additive manufacturing, FEM simulation using MATLAB, Computational fluid dynamics simulation, Thermo-mechanical simulation, Process simulation, Performance simulation & Uncertainty Quantification.

Projects: Evaluation of Thermomechanical (Process & Performance simulation) fabricated by the Laser-Powder Bed Fusion process, Design and analysis for oral maxillofacial surgery and depiction of fracture mechanics of the tibia bone for patients using CT Scan files and design for additive manufacturing for medical applications to treat patients (human tibia and skull).

PQE Group — Consultant **Sep 2021 – Mar 2022**

- Delivered 20 manufacturing and machine-learning projects, managed a 10-member team, prepared delivery plans, and coordinated with IT stakeholders to meet project timelines and KPIs.

Robert Bosch R&D — Project Trainee **Feb 2020 – Jul 2020**

- Analysed linear and nonlinear stability behaviour in connected-vehicle systems and developed microcontroller logic to coordinate human-driven and autonomous operation.

RESEARCH, TEACHING & TECHNICAL LEADERSHIP

Massachusetts Institute of Technology Science Policy Review — Technology Director **Oct 2024 – Feb 2026**

- Directed technology initiatives at the intersection of generative AI and public policy, integrating LLM workflows to support evidence synthesis, editorial processes, and decision-making. Published research on equitable, ethical, and fair data and AI governance for large language models (https://sciencepolicyreview.pubpub.org/pub/xk9f6tzw/release/4)

Stanford University — Machine Learning Specialist **Sep 2024 – Nov 2024**

- Developed physics-informed and generative machine-learning workflows for physical science and environmental applications using Stanford HPC and Google Cloud TPUs. Published Convolutional LSTM and PredRNN research for spatiotemporal modelling in 2024 (https://www.researchgate.net/publication/384665581_Convolutional_LSTM_PredRNN_Model)

Harvard Medical School — Project Trainee & Research Collaborator **Mar 2022 – May 2024**

- Optimised coaxial microfluidic bioprinting flow conditions to produce stable single- and multi-walled structures and improve structural stability by approximately 30%. Applied genetic algorithms, uncertainty quantification, finite-element modelling, and machine learning to molecular flow, absorption, blood-flow, and multilayer-wrinkling problems.

Publication: https://spiral.imperial.ac.uk/entities/publication/c1f9b71e-13cb-416f-bfdd-8834599d8508

Imperial College London — Researcher / Graduate Teaching & Research Assistant **2022 – 2024**

- Work focuses on gathering high-fidelity and low-fidelity numerical simulations data using Nektar++ (Solver based on Applied Mathematics).The utilization of the higher polynomial distribution in calculating the Coefficient of lift and drag has demonstrated superior accuracy and precision. Further, Co-kriging Data fusion and Adaptive sampling technique has been used to obtain the precise data predictions for the lift and drag within the confined domain (for the angle of attack ranging from 1 to 7 degree) without conducting the costly simulations on HPC clusters. This creates a methodology to quantifying uncertainty in CFD by minimizing the required number of samples. Developed multi-fidelity deep neural networks for 1D, 32D, and 100D uncertainty-propagation benchmarks, outperforming Co-Kriging in high-dimensional settings and accelerating training on an NVIDIA Tesla K80 GPU.

- Combined Nektar++, adaptive sampling, Bayesian optimisation, data fusion, and CFD simulations to predict NACA0012 lift and drag while reducing reliance on expensive high-fidelity samples.
- Taught and assessed machine learning, digital health, optimisation, deep learning, clustering, classification, generative AI, and related applied-computing topics.

Oxford Machine Learning (University of Oxford) — Graduate Teaching Assistant **2025 – 2025**

- I have worked as a teaching assistant to teach Machine Learning, Artificial Intelligence, Generative AI, Application of Machine Learning in Health and Biology, Representation Learning, Presentation learning, Agentic AI, human + AI alignment, Building GenAI products, Foundation models, large frontier models in applied domains, DNN, CNN, Optimisation, Deep learning, Clustering & Classification.

EDUCATION

Imperial College London — *London, UK* **Graduated Sep 2024**

Master's Degree (Intense Focus: Applied Mathematics & Machine Learning)

- Published the thesis "Uncertainty Quantification for Multi-Fidelity Simulations" and completed professional-level training in computing and machine learning. https://arxiv.org/abs/2503.08408

Research Focus: Applied Mathematics, Scientific Computing, Machine Learning & Quantum Computing

Accomplishment: Staff Recognition for Teaching and Supervision, Graduate Teaching & Research Assistant

Application Evaluator - UK Commonwealth Startup Fellowship

Affiliate-Grantham Institute & Imperial President Award Nominee (Societal Engagement & Supporting Student Experience)

Bühler Project-AI-Driven Microbial Growth Kinetics for Bioprocess Optimisation (Bayesian Hyperparameter Optimisation)

Imperial College, Department of Computing & Business School (Machine Learning & Artificial Intelligence), First-Class

ETH Zurich — *Zurich, Switzerland* **Completed Oct 2022**

Visiting Graduate Student — Robotics, Control Systems, Reinforcement Learning & Ethics

Research: Design, Modeling, Model predictive control, Control, Reinforcement learning & Trajectory Optimization.

Project: Design evaluation and performance optimization of an all-wheel drive miniature car capable of torque vectoring

- Led control-systems work for Swissloop Tunnelling and developed model predictive control, reinforcement-learning, and trajectory-optimization methods.

Vellore Institute of Technology & ARAI — *Vellore, India* **Graduated Sep 2020**

BTech Engineering — First Class, 91.3%, Department Rank 3

Department Rank – 3rd (VIT Vellore & ARAI) & 2nd Rank for Best Research – Bachelor's Thesis, Purdue University

Student of the Year Award (2019-2020) - SAE-India among all Indian Students & SAE-India Section Award (2019-2020)

Best Outgoing Student (Among Top 0.04% students out of 30,000) & AWS Course - Technical Essentials and Architecting

University of Oxford, Mathematical Institute — *Oxford, UK* **Completed August 2024**

Presentation learning, Agentic AI, human + AI alignment, Building GenAI products, Foundation models, and large frontier models in applied domains. Machine Learning and its applications in Health & Bio, and Representation Learning & Gen AI. Machine Learning, Factorization Methods, Clustering, Classification, Advanced Practical Topics in ML, DNN, Optimisation.

SELECTED PUBLICATIONS & RECOGNITIONS (Citation: 64, H-Index: 6. i-10 Index: 2)

- Swapnil Kumar, Quantifying Uncertainty in Quantum Machine Learning & Fluid mechanics, Princeton University & Imperial College London, 2025 (https://gss.pppl.gov/2025/Tuesday_flash.pdf)
- Reinforcement learning across temporal scales, Leila Wehbe, Swapnil Kumar, Patrick Mineault, Neuromatch Academy 2024, https://neuroai.neuromatch.io/tutorials/W1D2_ComparingTasks/student/W1D2_Tutorial3.html
- Swapnil Kumar, Sushila Maharjan, Valerio Luca Mainardi, Y. Shrike Zhang Microfluidics for Co-axial Bioprinting (Harvard Medical School), Imperial College London: https://spiral.imperial.ac.uk/handle/10044/1/108549
- https://compneuro.neuromatch.io/tutorials/W0D4_Calculus/student/W0D4_Tutorial2.html,
  https://compneuro.neuromatch.io/tutorials/W0D0_NeuroVideoSeries/student/W0D0_Tutorial3.html,
  https://compneuro.neuromatch.io/tutorials/W0D4_Calculus/student/W0D4_Tutorial3.html
- Nova 111 UK List 2026 for social impact, research, and education and contributed invited talks across the University of Oxford, University of Cambridge, Imperial College, Princeton, Tsinghua/GAUC, and related academic communities.
- Academic and Welfare Officer (PGT) - Faculty of Engineering, Imperial College London
- Lecturer - GAUC, Tsinghua University

{Lectures:} Scientific Machine Learning for Uncertainty Quantification, AI & machine learning applications in climate science, and Quantum computing for climate modeling.

- Boeing & Royal Aeronautical Society Scholarship for graduate studies at Imperial College London.
- Academic Judge - PhD Summer Showcase, Imperial College London
- Vedant Vashist, Neil Banthia, Swapnil Kumar, Prajwal Agrawal, A systematic review on materials, design, and manufacturing of swabs, Elsevier, https://www.sciencedirect.com/science/article/pii/S2666964122000467